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MS SQL Server Administration IIS404

Chapter Seven

Maintenance in MS SQL Server

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1. Managing Indexes

- لإدارة ال Indexes نقوم بقياس ال Fragmentation , حيث إن كان 30% أو أقل نقوم ب Reorganize

لل Index و في حال كان أكثر من 30% نقوم بعمل Rebuild لل Index

In order to manage indexes you have first to measure fragmentation by the following piece of code:

حيث نقوم بقياس ال Fragmentation باستخدام:

```
Select avg_fragmentation_in_percent, fragment_count from
sys.dm_db_index_physical_stats(DB_ID('DatabaseName'), OBJECT_ID('IndexName'),
1, null, 'Limited');
```

As best practice, if the fragmentation is about 30 you can reorganize the index, but if above it's recommended to rebuild it.

index will be rebuilt using the same columns, index type, uniqueness attribute, and sort order

index leaf level will be reorganized

Marks the index as disabled and unavailable for use by the Database Engine

```
ALTER INDEX {index_name | ALL}
ON [{database_name.[schema_name]. | schema_name.}]
{table_or_view_name}
{ REBUILD [WITH(<rebuild_index_option>[,...n])]
| REORGANIZE [ WITH( LOB_COMPACTION = {ON | OFF})]
| DISABLE
| SET (<set_index_option>[,...n]) }
```

حيث نقوم ب Rebuild في حال الداعي لذلك, كالتالي:

(Rebuild لكل جدول بجدوله)

```

ALTER INDEX ALL ON [Forewin Go Live$Value Entry$VSIFT$6] Rebuild
ALTER INDEX ALL ON [Forewin Go Live$Value Entry$VSIFT$7] Rebuild
ALTER INDEX ALL ON [Forewin Go Live$Posted Whse_ Receipt Line] Rebuild
ALTER INDEX ALL ON [Forewin Go Live$Cust_ Ledger Entry] Rebuild
ALTER INDEX ALL ON [Forewin Go Live$Value Entry$VSIFT$11] Rebuild
ALTER INDEX ALL ON [Forewin Go Live$Value Entry$VSIFT$13] Rebuild
ALTER INDEX ALL ON [Forewin Go Live$Value Entry$VSIFT$14] Rebuild

```

133 % | Connected. (1/1) | SQL (15.0 RTM) | SQL\Administrator (84) | master | 00:00:00 | 0 rows

أو نقوم بالتعديل على جميع ال Tables دفعة واحدة:

```
exec sp_msforeachtable 'alter index all on ? rebuild'
```

- ينصح القيام بعملية ال Index Rebuild في فترات الليل بعد الدوام، حيث سينخفض الأداء كثيرا بسبب إستهلاك الموارد في هذه العملية

2. Check Database Integrity Task

- يتم القيام بعملية تحقق من ال Database Integrity باستخدام تعليمة ال DBCC

The Check Database Integrity task checks the allocation and structural integrity of all the objects in the specified database. The task can check a single database or multiple databases, and you can choose whether to also check the database indexes.

The Check Database Integrity task encapsulates the DBCC CHECKDB statement.

3. Shrink a Database

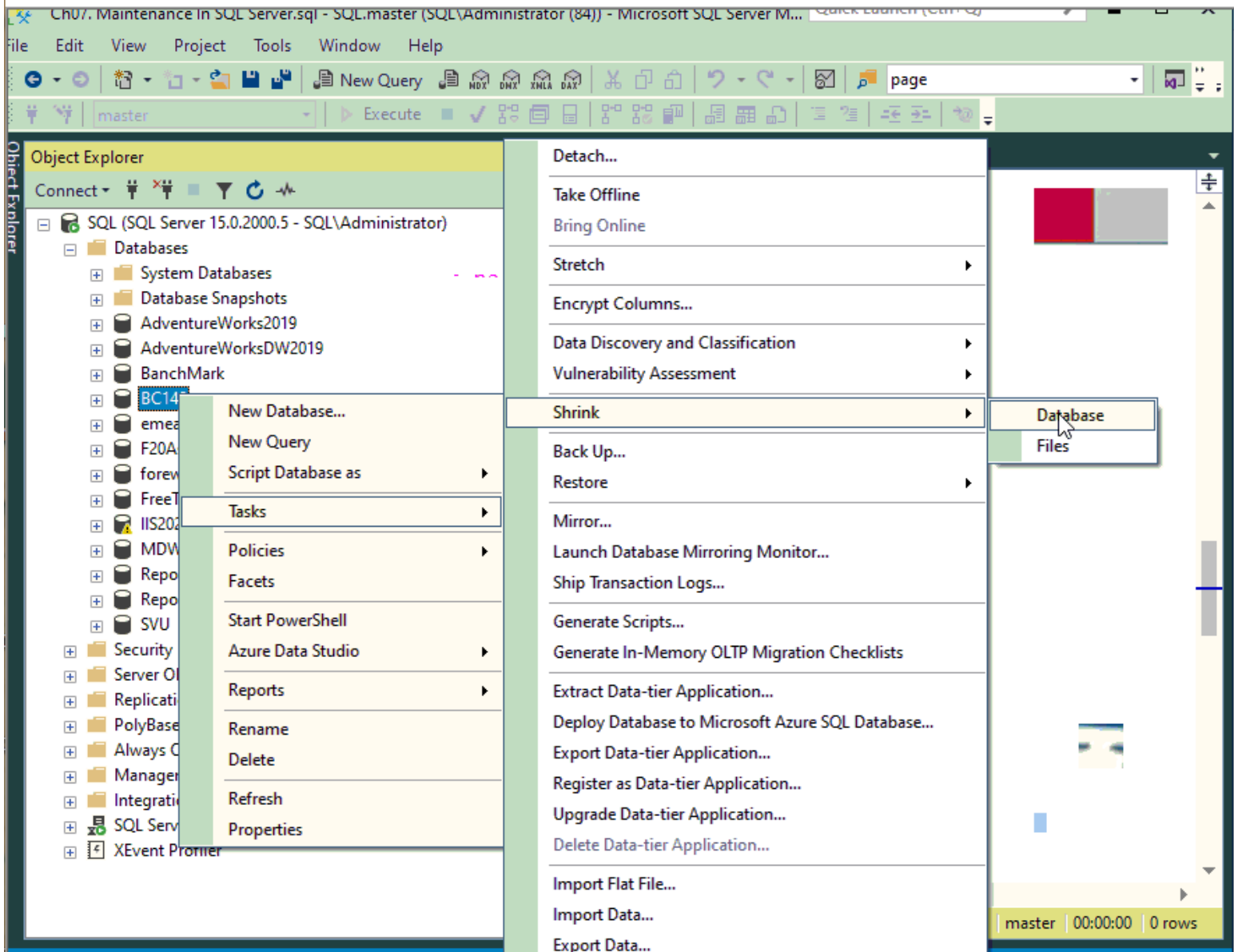
- عملية ال Shrinking لقاعدة البيانات تهدف لتقليص مساحتها المستخدمة

- النقطة السلبية الخاصة بها, أنه عند تقليص قاعدة البيانات يتم زيادة ال **Fragmentation**

داخل ال Indexes

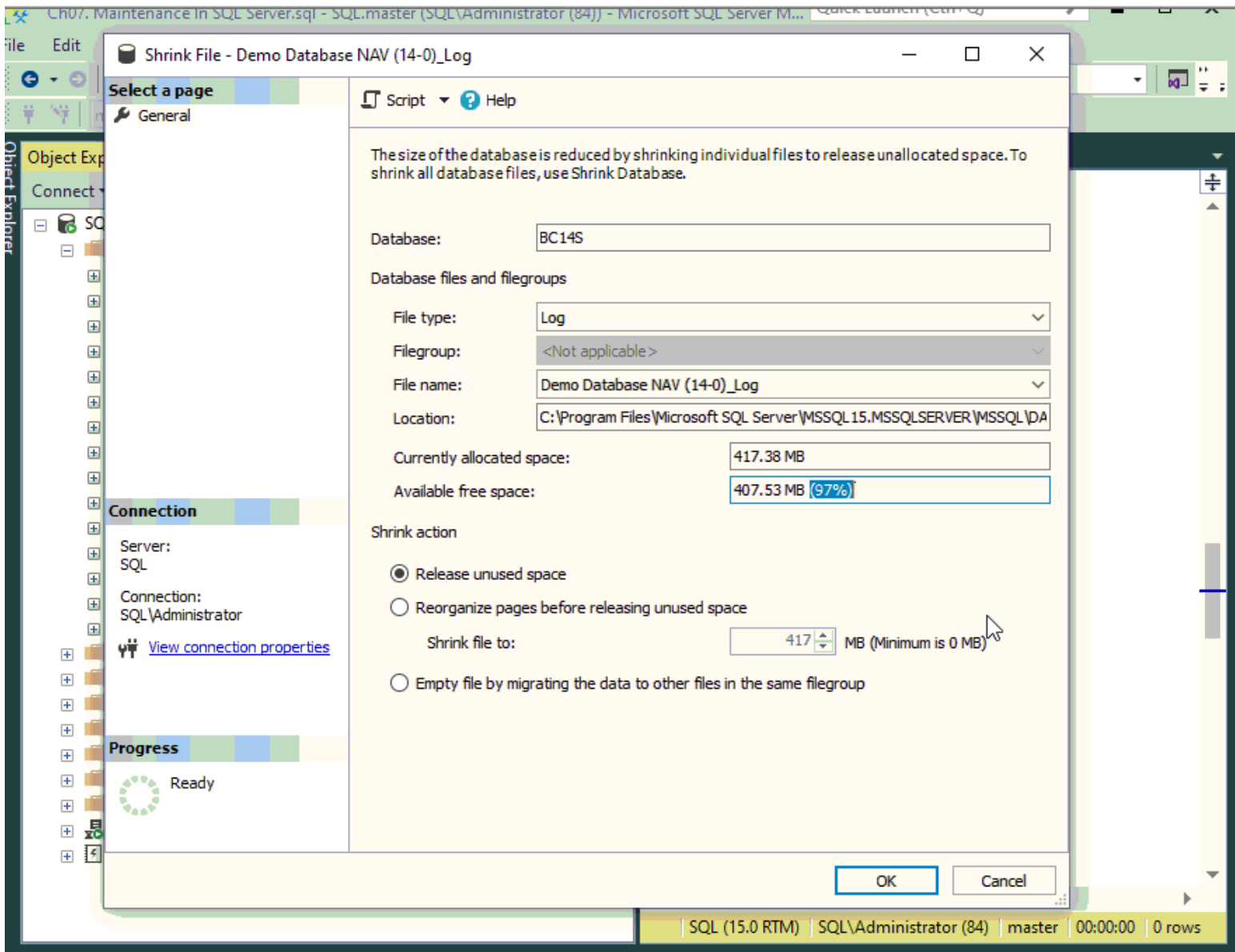
لذلك يجب علينا المقايضة بين تقليص الحجم, و زيادة ال **Fragmentation**

و للقيام بها نتبع الخطوات التالية:



Shrinking data files recovers space by moving pages of data from the end of the file to unoccupied space closer to the front of the file. When enough free space is created at the end of the file, data pages at end of the file can be deallocated and returned to the file system.

- و يكون أغلب النفع عند تطبيق ال Shrinking على ملفات ال Log, و يتم عرض الحجم الذي سيتحرر لدينا عند إختيار ملف معين أو قاعدة البيانات بأكملها:



3.1. Limitations and Restrictions

- القيود لعملية ال Shrinking:

(1) لا يمكن القيام ب *Backup* عند القيام ب *Shrinking* أو العكس

(2) لا يمكن تقليص حجم ال *DB* لأقل من الحجم المضبوط عند إنشائها , أو الحجم المضبوط لها باستخدام تعليمة *DBCC*

- The database cannot be made smaller than the minimum size of the database. The minimum size is the size specified when the database was originally created, or the last explicit size set by using a file-size-changing operation, such as *DBCC SHRINKFILE*. For example, if a database was originally

created with a size of 10 MB and grew to 100 MB, the smallest size the database could be reduced to is 10 MB, even if all the data in the database has been deleted.

- You cannot shrink a database while the database is being backed up. Conversely, you cannot backup a database while a shrink operation on the database is in process.

3.2. Recommendations

- نصائح يجب أخذها بعين الاعتبار عند القيام ب **Shrinking**:

(1) ينصح بالقيام ب **Shrining** بعد عملية تستهلك مساحة كبيرة , مثل **Truncate table, drop table** أو حذف أي كمية كبيرة من ال **Data**

(2) القيام بعملية **Shrink** بشكل متكرر خلال فترات متقاربة يعتبر مضيعة للوقت, حيث أن ال **DB** تحتاج لمساحة للعمليات اليومية

(3) ينصح بعدم تفعيل خيار ال **Auto_Shrink** لل **Database** إلا في حال لدينا سبب قوي

(4) يجب تذكر أن عمليات ال **Shrinking** تزيد من ال **Fragmentation** في ال **Database**

- To view the current amount of free (unallocated) space in the database:

```
SELECT file_id, name, type_desc, physical_name, size, max_size
FROM sys.database_files ;
```

- Consider the following information when you plan to shrink a database:
 - A shrink operation is most effective after an operation that creates lots of unused space, such as a truncate table or a drop table operation.
 - Most databases require some free space to be available for regular day-to-day operations. If you shrink a database repeatedly and notice that the database size grows again, this indicates that the space that was shrunk is required for regular operations. In these cases, repeatedly shrinking the database is a wasted operation.
 - A shrink operation does not preserve the fragmentation state of indexes in the database, and generally increases fragmentation to a degree. This is another reason not to repeatedly shrink the database.
 - Unless you have a specific requirement, do not set the AUTO_SHRINK database option to ON.

4. Update statistics

- هي معلومات يخرنها ال **query Optimizer** لإستخدامها في رسم و إعطاء ال **Query Execution Plan** الأفضل

- يفضل القيام بها في أيام العطل و في فترات متباعدة لأنها تستهلك الكثير من الموارد و تقوم بتشغيل ال **tempdb** بشكل كثير

- نستطيع القيام ب **Update Statistics** ل:

(1) **Object** معين

(2) **Table** معين

(3) **Database** بشكل كامل

Statistics refers to the statistical information about the distribution of values in one or

more columns of a table or an index. The SQL Server Query Optimizer uses this statistical information to estimate the cardinality, or number of rows, in the query result to be returned, which enables the SQL Server Query Optimizer to create a high-quality query execution plan. For example, based on these statistical information SQL Server Query Optimizer might decide whether to use the index seek operator or a more resource-intensive index scan operator in order to provide optimal query performance. In this article series, I am going to talk about statistics in detail.

You can update query optimization statistics on a table or indexed view in SQL Server 2019 (15.x) by using SQL Server Management Studio or Transact-SQL. By default, the query optimizer already updates statistics as necessary to improve the query plan; in some cases you can improve query performance by using UPDATE

STATISTICS or the stored procedure `sp_updatestats` to update statistics more frequently than the default updates.

Updating statistics ensures that queries compile with up-to-date statistics. However, updating statistics causes queries to recompile. We recommend not updating statistics too frequently because there is a performance tradeoff between improving query plans and the time it takes to recompile queries. The specific tradeoffs depend on your application. `UPDATE STATISTICS` can use `tempdb` to sort the sample of rows for building statistics.

4.1. *Permissions*

If using `UPDATE STATISTICS` or making changes through SQL Server Management Studio, requires `ALTER` permission on the table or view. If using `sp_updatestats`, requires membership in the `sysadmin` fixed server role, or ownership of the database (`dbo`).

4.2. *Using SSMS*

- In Object Explorer, click the plus sign to expand the database in which you want to update the statistic.
- Click the plus sign to expand the Tables folder.
- Click the plus sign to expand the table in which you want to update the statistic.
- Click the plus sign to expand the Statistics folder.
- Right-click the statistics object you wish to update and select Properties.
- In the Statistics Properties -statistics_name dialog box, select the Update statistics for these columns check box and then click OK.

4.3. *Using T-SQL*

4.3.1. To update a specific statistics object

The following example updates the statistics for the `AK_SalesOrderDetail_rowguid` index of the `SalesOrderDetail` table.

```
UPDATE STATISTICS Sales.SalesOrderDetail AK_SalesOrderDetail_rowguid;
```

4.3.2. To update all statistics in a table

The following example updates the statistics for all indexes on the `SalesOrderDetail` table.

```
UPDATE STATISTICS Sales.SalesOrderDetail;
```

4.3.3. To update all statistics in a database:

```
EXEC sp_updatestats;
```

5. History Cleanup Task

- هناك العديد من العمليات مثل عمليات ال Backup سوءا المأتمنة أم اليدوية, تقوم بترك ملفات في ال History Table الخاص بال msdb مثل ال ...logs حيث يقوم ال SQL Server بالتخلص منهم بشكل دوري, ولكن في حال أردنا التخلص منهم بشكل يدوي نستطيع إستخدام أداة ال History Cleanup - نستطيع أتمتة إستخدام هذه الأداة, و يفضل إستخدامها في أيام العطلة -

The History Cleanup task deletes entries in the following history tables in the SQL Server msdb database.

- backupfile
- backupfilegroup
- backupmediafamily
- backupmediaset
- backupset
- restorefile
- restorefilegroup
- restorehistory

By using the History Cleanup task, a package can delete historical data related to backup and restore activities, SQL Server Agent jobs, and database maintenance plans.

This task encapsulates the sp_delete_backuphistory system stored procedure and passes the specified date to the procedure as an argument.

6. Fixing Database States

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A database is always in one specific state. For example, these states include ONLINE, OFFLINE, or SUSPECT. To verify the current state of a database, select the state_desc column in the sys.databases catalog view or the Status property in the DATABASEPROPERTYEX function.

6.1. DATABASE STATE DEFINITIONS

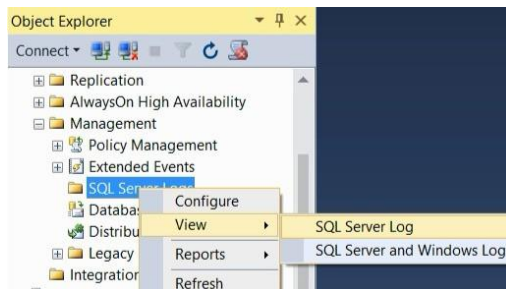
State	Definition
ONLINE	هي الحالة الطبيعية لل DB التي تكون بها فعالة Database is available for access. The primary filegroup is online, although the undo phase of recovery may not have been completed.
OFFLINE	يمكن أن تصبح بهذه الحالة نتيجة استخدام يدوي لتعليمة: (لكي نعدّل مكان ملف معين منها) Alter database database_name set offline أو نتيجة لعملية backup أو عملية أخرى.. و تصبح بها ال DB غير قابلة للاستخدام Database is unavailable. A database becomes offline by explicit user action and remains offline until additional user action is taken. For example, the database may be taken offline in order to move a file to a new disk. The database is then brought back online after the move has been completed.

RESTORING	توضع بها ال DB عند القيام ب restore لها أو لبعض الملفات منها مع خيار ال With Norecovery , حيث تصبح غير قابلة للإستخدام و نقوم بعملية ال Restore بدون أن ندخل إلى ال DB (بدون أن نستخدم use database database_name) و أخيراً عند إستخدام ال Restore مع خيار ال With Recovery تعود ال DB لوضع Offline One or more files of the primary filegroup are being restored, or one or more secondary files are being restored offline. The database is unavailable.
RECOVERING	هي حالة مؤقتة تعود بعدها ال DB إلى وضع Online في حال نجاح عملية ال Recovery و في حال عدم نجاح عملية ال Recovery ستصبح ال DB غير قابلة للإستخدام توضع بها ال DB في حال البدء بعملية معينة و توقفها بنصف العملية بسبب إستهلاك جميع الموارد أو بسبب إعادة إقلاع ال Services نتيجة عملية معينة أو بسبب إنقطاع كهرباء مفاجئ عن ال DB, حيث سيتم عند إعادة الإقلاع قيام ال Server بعملية Crash Recovery (إختيار إما عمل Redo أو Undo للمناقشات في ال Log) و عادة تأخذ عملية ال Recovery وقت طويل و تكون بها ال DB غير قابلة للإستخدام Database is being recovered. The recovering process is a transient state; the database will automatically become online if the recovery succeeds. If the recovery fails, the database will become suspect. The database is unavailable.
RECOVERY PENDING	تصبح ال DB في هذه الحالة, في حال البدء بعملية Recovery و عدم القدرة على إنهائها بسبب نفاذ الموارد (نفاذ ال RAM, أو المساحة التخزينية ..) و تكون بها ال DB غير قابلة للإستخدام SQL Server has encountered a resource-related error during recovery. The database is not damaged, but files may be missing or system resource limitations may be preventing it from starting. The database is unavailable. Additional action by the user is required to resolve the error and let the recovery process be completed.
SUSPECT	هذه الحالة تكون نتيجة عطل أو ضياع لل Primary File Group , و لا يمكن إسترجاعها أو إستخدامها في هذا الحالة At least the primary filegroup is suspect and may be damaged. The database cannot be recovered during startup of SQL Server. The database is unavailable. Additional action by the user is required to resolve the problem.
EMERGENCY	هي حالة يقوم المستخدم بتفعيلها يدوياً, تصبح بها ال DB في وضع Single-User Mode و تصبح في حالة Read-Only, و لا يمكن تسجيل الدخول عليها و يصبح جانبها مثلث أصفر و تحصر جميع الصلاحيات بال sys admins فقط (لذلك الذي يقوم بتفعيل هذه الحالة يجب أن يكون من أعضاء ال sys admins) و تستخدم هذه الحالة في ال Troubleshooting , كإستخدامها لإخراج ال DB من وضع Suspect مثلاً لتفعيل هذه الحالة نستخدم: Alter database database_name set emergency User has changed the database and set the status to EMERGENCY. The database is in single-user mode and may be repaired or restored. The database is marked READ_ONLY, logging is disabled, and access is limited to members of the sysadmin fixed server role. EMERGENCY is

primarily used for troubleshooting purposes. For example, a database marked as suspect can be set to the EMERGENCY state. This could permit the system administrator read-only access to the database. Only members of the sysadmin fixed server role can set a database to the EMERGENCY state.

Date ▾	Source	Message
06.09.20 2:52:25 PM	spid5s	Recovery is complete. This is an informational message only. No user action is required.
06.09.20 2:52:25 PM	spid5s	Recovery completed for database DF (database ID 9) in 1079 second(s) (analysis 3594 ms, redo 398873 ms, undo 673500 ms.) This is an informational message only. No user action is required.

When your database sticks in recovery, then you have to wait and monitor via SQL Server log.



- + System Databases
- + Database Snapshots
- + ChangeLog
- + DF (In Recovery)
- + Dutyfree
- + HQ
- + Shadow

Log file summary: No filter applied		
Date ▾	Source	Message
06.09.20 2:39:48 PM	spid24s	Recovery of database 'DF' (9) is 1% complete (approximately 31930 seconds remain). Phase 2 of 3. This is an informational message only. No user action is required.
06.09.20 2:39:29 PM	spid24s	Recovery of database 'DF' (9) is 0% complete (approximately 38707 seconds remain). Phase 2 of 3. This is an informational message only. No user action is required.

7. Procedure to Recover SQL Database from SUSPECT Mode



Sometimes, when connecting to an SQL Server, you may find that the SQL database (db) is marked as 'SUSPECT'. This may happen due to several reasons like missing or corrupt transactional log file of the database, faulty hardware, virus attack, abrupt shutdown of SQL server, etc. View the SQL Server error log to know the specific cause behind SQL Server database suspect mode.

- هناك عدة خطوات تستخدم لعمل Recovery من حالة Suspect, حيث في حال عدم نجاح أي منها نحاول بالتالي تلبيها بالتسلسل:

When SQL Database goes in **suspect mode**, you cannot access the db and cannot perform any transactions until it is back online. In order to resolve this problem, follow these steps:

1. Try to bring the database back online using the following line of code, but if this does not work, as expected, then move to point 2:

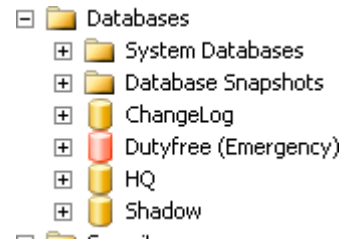
```
ALTER DATABASE DBName SET ONLINE
```

2. Try DBCC CHECKDB as the following

```
DBCC CHECKDB ( 'DBName' , REPAIR_ALLOW_DATA_LOSS ) WITH  
ALL_ERRORMSG, NO_INFOMSGS;
```

3. Sometimes you get error trying point 2, so try to set the database in **Emergency** mode, but setting database in Emergency.

```
ALTER DATABASE dutyfree SET EMERGENCY
```



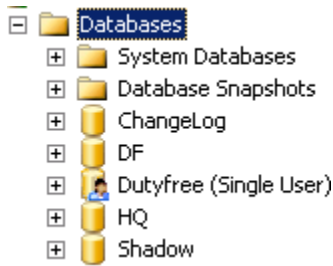
4. Sometimes you get the error while trying to set the database in Emergency mode, because the database is locked, so whenever you face a lock in the database while maintenance search for the reason session and kill it then proceed point 3.

```
EXEC sp_who2
```

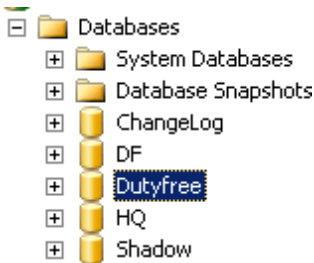
```
kill 58 (suppose the session is 58)
```

4. then if you try point 2, an error message demands putting the database in single user mode before executing this command so you have to execute the following:

```
Alter database DBName set single_user
```



Then proceed point 2:



Summary

```

EXEC sp_resetstatus 'dutyfree';
ALTER DATABASE dutyfree SET EMERGENCY

--01
ALTER DATABASE dutyfree SET ONLINE --fail?
--02
DBCC CHECKDB ('Dutyfree', REPAIR_ALLOW_DATA_LOSS) WITH ALL_ERRORMSG, NO_INFOMSGS; --fail?
--03
ALTER DATABASE dutyfree SET EMERGENCY --fail?
--04
EXEC sp_who2
kill 58
kill 59
kill 61
kill 51
kill 53
--05
ALTER DATABASE dutyfree SET EMERGENCY
--06
alter database dutyfree set single_user
DBCC CHECKDB ('Dutyfree', REPAIR_ALLOW_DATA_LOSS) WITH ALL_ERRORMSG, NO_INFOMSGS;
alter database dutyfree set multi_user

```

Following the above steps will likely help you fix SQL database suspect mode issue. If the problem still persists, restore the database from clean and updated backup that you created before the **database in suspect mode** problem. Restoring db from backup is the best and fastest method to fix the problem.

8. Maintenance Plans

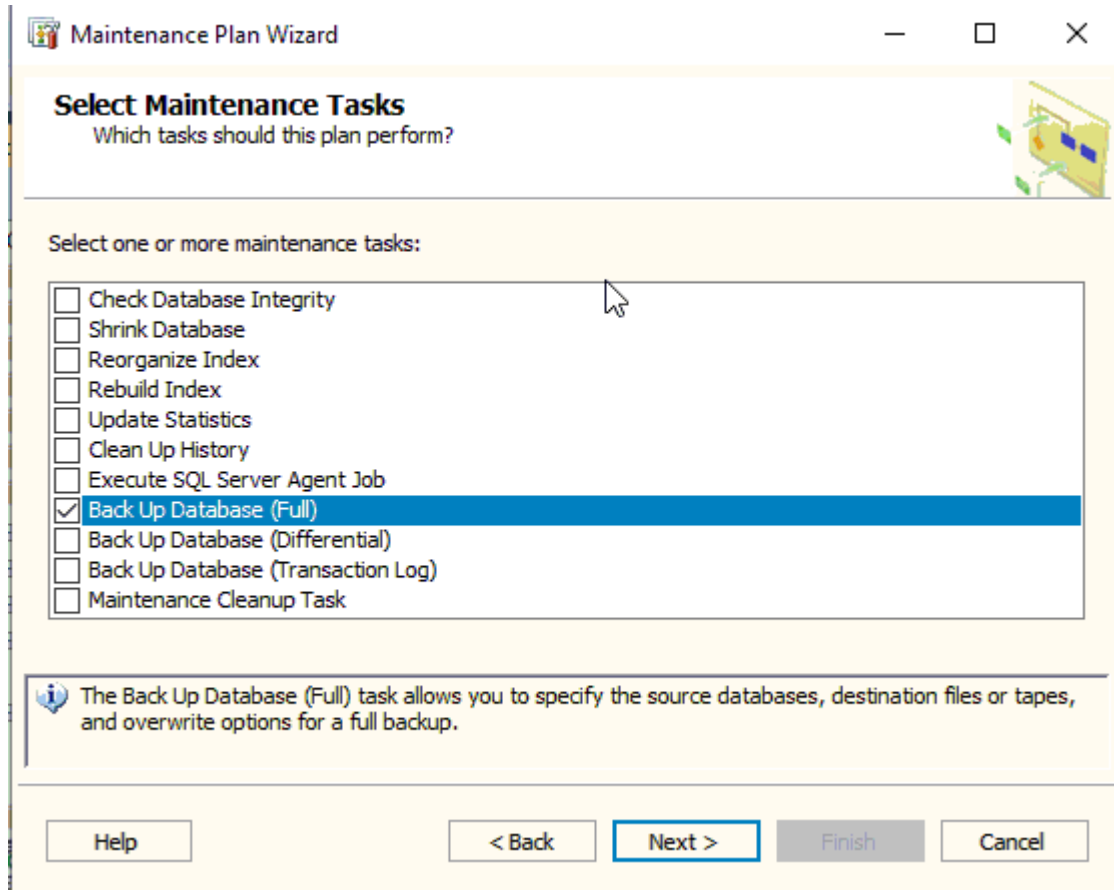
Maintenance plans create a workflow of the tasks required to make sure that your database is optimized, regularly backed up, and free of inconsistencies.

SQL Server 2017 maintenance plans provide the following features:

- ال Maintenance Plans هي مهام مؤتمتة نقوم بإنشائها من ال Agent عن طريق

ال SQL Server Maintenance Wizard , ونعطيها اسم يكون مفهوم لطبيعة عملها

ثم جدولة أوقات تنفيذها, ثم نحدد أحد ال Maintenance tasks التي قمنا بدراستها منذ قليل.



ثم نحدد ال DB او عدة DBs التي نريد تطبيق هذه ال Maintenance plan عليها

- Workflow creation using a variety of typical maintenance tasks. You can also create your own custom Transact-SQL scripts.

- Conceptual hierarchies. Each plan lets you create or edit task workflows. Tasks in each plan can be grouped into subplans, which can be scheduled to run at different times.
- Support for multiserver plans that can be used in master server/target server environments.
- Support for logging plan history to remote servers.
- Support for Windows Authentication and SQL Server Authentication. When possible, use Windows Authentication.

Maintenance plans can be created to perform the following tasks:

- Reorganize the data on the data and index pages by rebuilding indexes with a new fill factor. Rebuilding indexes with a new fill factor makes sure that database pages contain an equally distributed amount of data and free space. It also enables faster growth in the future.
- Compress data files by removing empty database pages.
- Update index statistics to make sure the query optimizer has current information about the distribution of data values in the tables. This enables the query optimizer to make better judgments about the best way to access data, because it has more information about the data stored in the database. Although index statistics are automatically updated by SQL Server periodically, this option can force the statistics to update immediately.
- Perform internal consistency checks of the data and data pages within the database to make sure that a system or software problem has not damaged data.
- Back up the database and transaction log files. Database and log backups can be retained for a specified period. This lets you create a history of backups to be used if you have to restore the database to a time earlier than the last database backup. You can also perform differential backups.
- Run SQL Server Agent jobs. This can be used to create jobs that perform a variety of actions and the maintenance plans to run those jobs.
- The results generated by the maintenance tasks can be written as a report to a text file or to the maintenance plan tables (sysmaintplan_log and

sysmaintplan_logdetail) in msdb. To view the results in the log file viewer, right-click Maintenance Plans, and then click View History.

9. References

<https://docs.microsoft.com/en-us/sql/relational-databases/databases/database-states?view=sql-server-ver15>

<https://docs.microsoft.com/en-us/sql/relational-databases/databases/shrink-a-database?view=sql-server-ver15>

<https://docs.microsoft.com/en-us/sql/relational-databases/statistics/update-statistics?view=sql-server-ver15>

<https://docs.microsoft.com/en-us/sql/integration-services/control-flow/history-cleanup-task?view=sql-server-ver15#:~:text=The%20History%20Cleanup%20task%20deletes,the%20SQL%20Server%20msdb%20database.&text=By%20using%20the%20History%20Cleanup.jobs%20C%20and%20database%20maintenance%20plans.>